

Flood Susceptibility Model

Model #3 of the AquaGuard / Bangladesh Flood Early-Warning System

A static, terrain-based hazard model — literature review, methodology, real data, and evaluation results

August 14, 2026

Companion to the flood-risk-classifier and discharge-forecaster reports

1. Introduction & Purpose

This project already includes two temporal flood-forecasting models: a flood-risk classifier (flood / no-flood probability at 24h/48h/72h, driven by live rainfall/soil-moisture/discharge data) and a discharge forecaster (predicted river flow in m³/s at the same horizons). Both answer variants of the same question — *will conditions here become dangerous soon* — and both are honestly hard problems: rare, sparse positive labels, and precision that stays modest even at a carefully tuned recall target.

This report covers a third, deliberately different model: **flood susceptibility** — a static assessment of how flood-prone a piece of ground is by its own geography (elevation, slope, distance to the nearest river, drainage density, land cover), independent of any particular storm. It does not compete with the other two models; it answers a question they cannot: even before this year's weather is known, is this specific patch of land the kind that floods, or the kind that stays dry?

The three models combine into one pipeline: the classifier's time-bound hazard probability, modulated by this model's static susceptibility score, via a transparent formula documented in Section 9 — deliberately not a black-box stacked meta-model, so the combination stays explainable in one line.

2. Literature Review

A multi-pass literature search (methodology, Bangladesh-specific studies, data-source accuracy, and evaluation-methodology papers — roughly 20 separate search queries, not a single pass) was conducted before design work began. The sources below are the ones that materially shaped this model's design, grouped by theme.

2.1 General methodology & model-comparison studies

Source	Key finding relevant to this project
Flood susceptibility assessment using three ML techniques and comparison of their performance (<i>Scientific Reports</i> , 2026)	Random Forest, Gradient Boosting, and XGBoost all outperform simpler models on nonlinear terrain relationships; ensemble trees are the consistent starting point across the field.
Integrating geospatial intelligence and machine learning for flood susceptibility mapping (<i>Scientific Reports</i>)	Reinforces elevation, slope, and distance-to-river as consistently top-ranked predictors across independent studies.
Flood susceptibility mapping using supervised ML models: insights into predictors' significance and models' performance (<i>Geocarto International</i>)	Systematic predictor-importance comparison across model families — informed this project's choice to report SHAP importance rather than a single opaque accuracy number.
Combining ML Models and Satellite Data of an Extreme Flood Event for Flood Susceptibility Mapping (<i>MDPI Water</i>)	Uses satellite-observed flood extent directly as training labels — the same core idea this project used with the Copernicus GFM SAR archive, aggregated across time instead of one event.
Enhancing flood susceptibility mapping through nature-inspired metaheuristic algorithms and ML (<i>ScienceDirect</i>)	RNN-GA / LSTM-GA hybrids reach ~93% AUC but require metaheuristic optimization overhead judged out of scope for this project's data volume (30 stations, not thousands of gauges).
Next generation data-driven flood susceptibility modelling with spatial machine learning (<i>ScienceDirect</i>)	Directly informed this project's evaluation design: documents that random train/test splits on spatially autocorrelated data inflate reported AUC by 5–15%.
Spatial validation reveals poor predictive performance of large-scale ecological mapping models (<i>PMC</i>)	Cross-domain (ecology, not hydrology) confirmation of the same random-vs-spatial-split inflation problem — the issue is general to spatial ML, not specific to flood studies.

2.2 Bangladesh-specific studies

Source	Key finding relevant to this project
Predictive Analytics for Floods in Bangladesh: A Comparative Exploration of ML and Deep Learning Classifiers (<i>ICCIT 2023 / IEEE</i>)	LightGBM (97.76%) and Random Forest (97.71%) led 10 compared classifiers on 65 years of Bangladesh data — but on a different task (see §7 comparison) and evaluation method than this project uses.
A machine learning-based approach for flash flood susceptibility mapping considering rainfall extremes, northeast Bangladesh (<i>ResearchGate</i>)	Random Forest outperformed XGBoost specifically in hilly, flash-flood-prone terrain — this project independently found the same result on its own hilly/flat mixed grid (§6.1).
Assessment and Zonation of Flood Susceptibility in Sylhet Division, Bangladesh, Using GIS and AHP (<i>Journal of Flood Risk Management</i> , 2025)	Non-ML (AHP, expert-weighted) baseline: ROC $\approx$ 87% — a useful, realistic reference point against inflated ML-only numbers.
GIS and AHP-based flood susceptibility mapping: a case study of Bangladesh (<i>Sustainable Water Resources Management</i>)	A second independent AHP baseline for Bangladesh, consistent with the Sylhet study above.
Geospatial driven machine learning approach for flood susceptibility mapping in southeastern Bangladesh (<i>Discover Geoscience</i> , 2026)	The most directly comparable regional ML study found during this search.

2.3 Data sources & terrain-analysis methodology

Source	Key finding relevant to this project
Vertical accuracy assessment of freely available global DEMs (FABDEM, Copernicus DEM, NASADEM, AW3D30, SRTM) in flood-prone environments (<i>International Journal of Digital Earth</i>)	Copernicus DEM GLO-30 (NMAD $\approx$ 1.27m) is substantially more accurate than SRTM ($\approx$ 3.65m), especially in flat terrain — directly determined this project's DEM choice.
TWI computation: a comparison of different open source GISs (<i>Open Geospatial Data, Software and Standards</i>)	Compared QGIS/SAGA/GRASS/Whitebox terrain toolchains — informed the choice of a pure-Python pipeline (pysheds) over a desktop-GIS dependency.
Suitability of the height above nearest drainage (HAND) model for flood inundation mapping in data-scarce regions (<i>Earth Science Informatics</i>)	HAND is specifically weaker in flat, low-relief, urbanized-drainage terrain — exactly Bangladesh's geography. This triggered a mid-build review of this project's own HAND usage (see §4.3).
HydroRIVERS v1.0 (Lehner & Grill, 2013, via HydroSHEDS)	A peer-reviewed, independently-built global river network (8.5 million reaches) — used in this project both as a validation cross-check for, and ultimately as the production source of, the distance-to-river feature (see §4.6).

3. Model Families Used in the Literature

Across every source reviewed, the following model families recur. This project's own choice (§4) is included in the same table for direct comparison of what each family is and how it performed elsewhere.

Model family	Where seen	Typical reported performance	Notes
Logistic Regression	Baseline in most comparative studies	Consistently the weakest of the compared models	Cannot capture nonlinear terrain interactions.
Random Forest	Very common; sometimes wins outright	97.71% accuracy (Bangladesh, §2.2); AUC 0.88–0.98 range across studies	Won in this project too (§6.1) — matches the hilly-terrain finding in §2.2.
XGBoost / Gradient Boosting	Very common	AUC up to 0.985 in some studies; beaten by RF in at least one Bangladesh hilly-terrain study	This project's classifier (model #1) also uses gradient boosting (LightGBM), for a different, temporal task.
LightGBM	Common, esp. recent Bangladesh studies	97.76% accuracy (Bangladesh, §2.2)	Compared head-to-head against RF for this model (§6.1); RF won on the spatial evaluation.
SVM / AdaBoost / ExtraTrees / MLP	Seen as comparison baselines	Generally mid-pack, below RF/XGBoost/LightGBM	Not tried in this project — the literature consensus already points to tree ensembles.
TabNet / TabPFN (deep tabular)	One Bangladesh comparative study	Competitive but not clearly better than LightGBM/RF	Deliberately not used here — added model complexity with no demonstrated benefit for this data volume (30 stations).
RNN-GA / LSTM-GA (deep learning + metaheuristic optimization)	One comparative study	~93% AUC, best-in-study	Requires substantially more data and optimization infrastructure than this project's scope justifies.
AHP (Analytic Hierarchy Process — expert-weighted, non-ML)	Multiple Bangladesh studies	ROC $\approx$ 87%	A realistic, non-inflated reference point (§7).
Random Forest (this project)	—	Spatial CV: 0.888 mean ROC-AUC; held-out test: 0.908 ROC-AUC, 0.785 PR-AUC	Chosen after a real head-to-head against LightGBM on this project's own data — not assumed from the literature.

4. Our Model: Design & Rationale

4.1 Why a third, static model

Models #1 and #2 both need live weather/discharge data and both forecast a moving target (this year's conditions). Susceptibility answers a fixed question instead — a genuinely different, complementary signal, not a third attempt at the same forecast. It also does not need live data at all: elevation, slope, and land cover do not change day to day, so once trained it can serve instantly without calling any external weather API.

4.2 The spatial grid

Rather than a single value per station (too coarse to train a spatial model on) or a uniform grid over all of Bangladesh (far more labeling effort than this project's 30-station scope needs), a local 7×7 grid (49 points, $\pm 0.09^\circ \approx 10\text{km}$ span) was built around each of the 30 monitored stations — 1,470 points in total. This directly serves the actual use case (scoring the neighborhoods this project already monitors) without the much larger labeling effort a national atlas would require.

4.3 A mid-build correction: reusing, not duplicating, existing terrain data

This project's temporal classifier (model #1) already includes two terrain features (elevation_m, hand_m) sourced from MERIT Hydro — a peer-reviewed, properly flow-routed Height Above Nearest Drainage (HAND) product, deliberately chosen over self-computing terrain metrics specifically to avoid a DEM-resampling mismatch bug (documented project decision log, §17–18). Early in this model's design, before checking, a self-computed replacement for those two features was nearly built — which would have both duplicated existing work and produced a strictly less rigorous version of an already-correct feature. This was caught before implementation. The two features actually built for this model (distance-to-river, drainage density) are genuinely new — they do not exist anywhere else in the project. Drainage density remains computed from a single, internally consistent DEM source end-to-end, a different situation from the cross-product resampling mismatch the original MERIT Hydro decision was avoiding. Distance-to-river started the same way, but was later revised after an independent cross-check (§4.6) — itself an example of the same "verify, don't assume" discipline applied a second time, this time to this project's own new feature rather than an existing one.

4.4 Model family selection

Rather than assume the classifier's own model choice (LightGBM) would transfer to a different problem, LightGBM and Random Forest were compared head-to-head on identical data via spatial cross-validation (§6.1). Random Forest won, consistent with a Bangladesh-specific finding (§2.2) that plain RF can outperform gradient boosting specifically in hilly terrain — this project's grid spans both flat delta and the hilly Chittagong Hill Tracts, so that distinction was worth checking rather than assuming.

4.5 Final hyperparameters

Parameter	Value
n_estimators	300
max_depth	8
min_samples_leaf	5
random_state	42

4.6 Post-hoc validation: cross-checking distance-to-river, and acting on the result

After the model above was built and evaluated, the self-derived distance-to-river feature was cross-checked against HydroRIVERS v1.0 (Lehner & Grill, 2013) — a peer-reviewed, independently built global river network, downloaded directly (91MB, no registration) and compared point-by-point against the pysheds flow-accumulation-derived distances. The two methods correlated poorly across the full 1,470-point grid (Pearson $r \approx 0.03$) — not the clean validation expected. At the 30 station centers specifically, both agreed on the right order of magnitude (a few hundred meters), with a consistent, explicable pattern: the self-derived method, built on a permissive flow-accumulation threshold over a fine 30m DEM, detects a dense network of minor channels and therefore always reports something at least as close as the coarser, major-rivers-only HydroRIVERS network. Away from station centers, the two increasingly measure different things — likely compounded by a known artifact of D8 flow routing on very flat deltaic terrain (spurious, overly dense drainage patterns).

Rather than assume either method was better, both were tried empirically: the full training pipeline (identical spatial CV, identical 7-station held-out test) was re-run three times, changing only the distance-to-river feature.

Variant	Spatial CV ROC-AUC	Held-out test ROC-AUC	Held-out test PR-AUC
Self-derived (pysheds) — original	0.892	0.902	0.770
HydroRIVERS — adopted	0.888	0.908	0.785
Both features together	0.889	0.903	0.773

HydroRIVERS won on the held-out test (the more decision-relevant metric) by a modest but real margin; combining both features performed no better than either alone, indicating the two carry mostly redundant signal rather than complementary information. Given the roughly-tied numbers plus the stronger methodological story (an externally validated, peer-reviewed product rather than a self-derived feature with a known flat-terrain algorithmic caveat), **the production model now uses the HydroRIVERS- based distance** — the figures and numbers throughout the rest of this report reflect that final, adopted version. The self-derived version is retained in the underlying data (`dist_to_river_m_pysheds_legacy`) for provenance, not deleted.

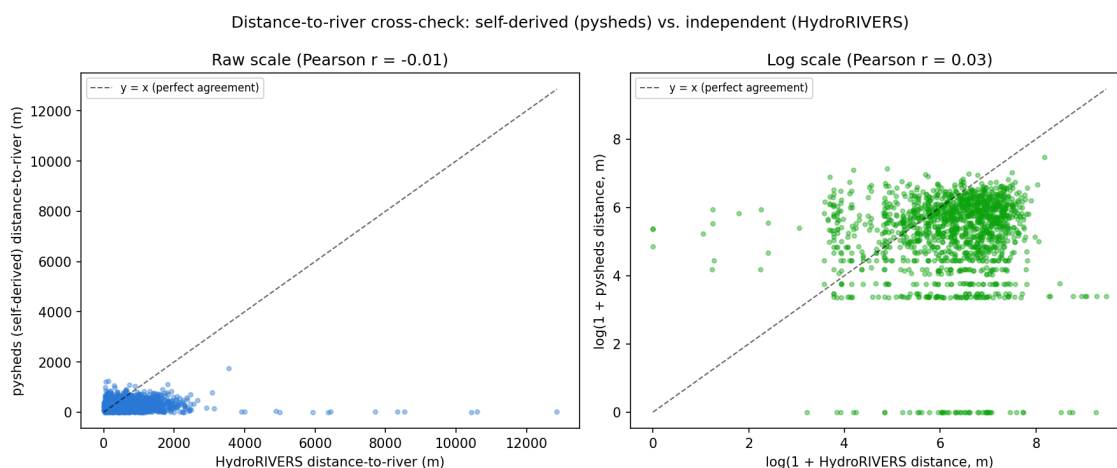


Figure 4.6.1. The cross-check that triggered this decision: self-derived vs. independent distance-to-river, both raw and log-scaled. Near-zero raw correlation led to the empirical comparison above, not a decision made on the plot alone.

5. Data Sources

Source	What it provides	Access
Copernicus DEM GLO-30 (ESA/Airbus, via AWS Open Data)	Elevation → slope, distance-to-river, drainage density (self-derived via flow-routing)	Free, unauthenticated, 1° Cloud-Optimized GeoTIFF tiles
ESA WorldCover v200 (2021)	10m land-cover class (11 categories: tree, cropland, water, wetland, mangrove, built-up, etc.)	Free, unauthenticated, 3° tiles, via AWS
Copernicus Global Flood Monitoring (GFM) — Sentinel-1 SAR	Flood observation labels, aggregated across the full 2016-04-01 to 2026-08-13 archive	Free, unauthenticated STAC catalog + Cloud-Optimized GeoTIFFs
MERIT Hydro v1.0.1 (referenced, not re-derived)	Peer-reviewed HAND/elevation values already used by model #1's STATIC_TERRAIN table	One-time Earth Engine sample (existing project asset, see §4.3)

5.1 Terrain feature derivation pipeline

Per 1° DEM tile (19 tiles cover the full 30-station grid): fill pits → fill depressions → resolve flats → D8 flow direction → flow accumulation (via pysheds, a pure-Python library — the initially-preferred richdem was rejected after a real failed installation: it requires a C++ compiler unavailable on the build machine). Slope is computed from a latitude-scaled elevation gradient (cell size varies with longitude at this latitude: $\approx 28\text{m} \times \approx 31\text{m}$, not square, despite the DEM's nominal “30m” label). “River” cells are flow-accumulation cells above 500 ($\approx 0.45\text{km}^2$ contributing area, a standard minimum stream-initiation threshold); drainage density is the river-cell count in a 1km-radius local window, divided by that window's area — this part of the pipeline is unchanged and still self-derived. **Distance-to-river is no longer taken from this pipeline's own Euclidean distance transform** — after a post-hoc cross-check against an independent river network (§4.6), it was replaced with a vector distance to the nearest HydroRIVERS reach, which measurably performed at least as well and rests on a peer-reviewed external dataset rather than a self-computed one. The original self-derived values are retained in the raw data for provenance.

5.2 Label construction

Every grid point was sampled against the full GFM archive: 1,470 points, 918,423 total valid satellite observations (median 575 per point). A point is labeled **flooded** if the archive ever recorded a detection there ($n_{\text{flooded}} \geq 1$); labeled **non-flooded** only if observed at least 200 times with zero detections — a deliberately conservative bar, chosen after inspecting the real observation-count distribution (median 575, well above the 200 threshold, so this bar excludes very little real data while still requiring substantial evidence). Points with fewer than 200 observations and zero detections are dropped as inconclusive, not assumed negative.

6. Example Data

Three real rows from the final training table (susceptibility_training_table.csv), illustrating the full feature set:

point_id	station	elevation (m)	slope (°)	dist. to river (m)	drainage density	land cover	label
DH01_0_2	DH01	29.3	4.94	1848	0.82	tree	non-flooded
GO01_6_2	GO01	11.9	3.63	1547	0.92	cropland	non-flooded
SW90_2_5	SW90	12.5	0.00	1003	2.08	water	non-flooded

Three real rows from the raw label-aggregation output (grid_point_flood_counts.csv), showing the observation depth behind each label:

point_id	station	basin	n_valid (observations)	n_flooded (detections)
CO01_4_2	CO01	ganges	578	0
OB01_5_0	OB01	brahmaputra	574	4
SW99_3_2	SW99	brahmaputra	575	0

7. Results: Plots & Analytics

Every figure below was regenerated directly from the trained model and the real data files on disk.

7.1 Data composition & coverage

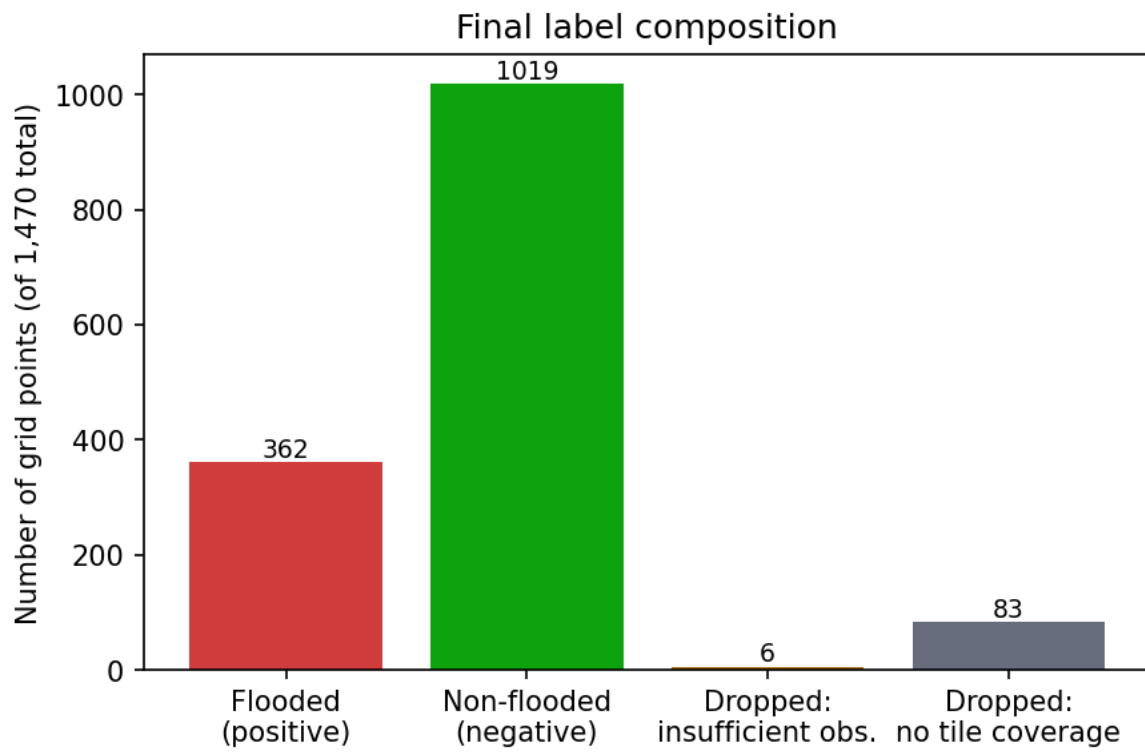


Figure 1. Final label composition across all 1,470 grid points.

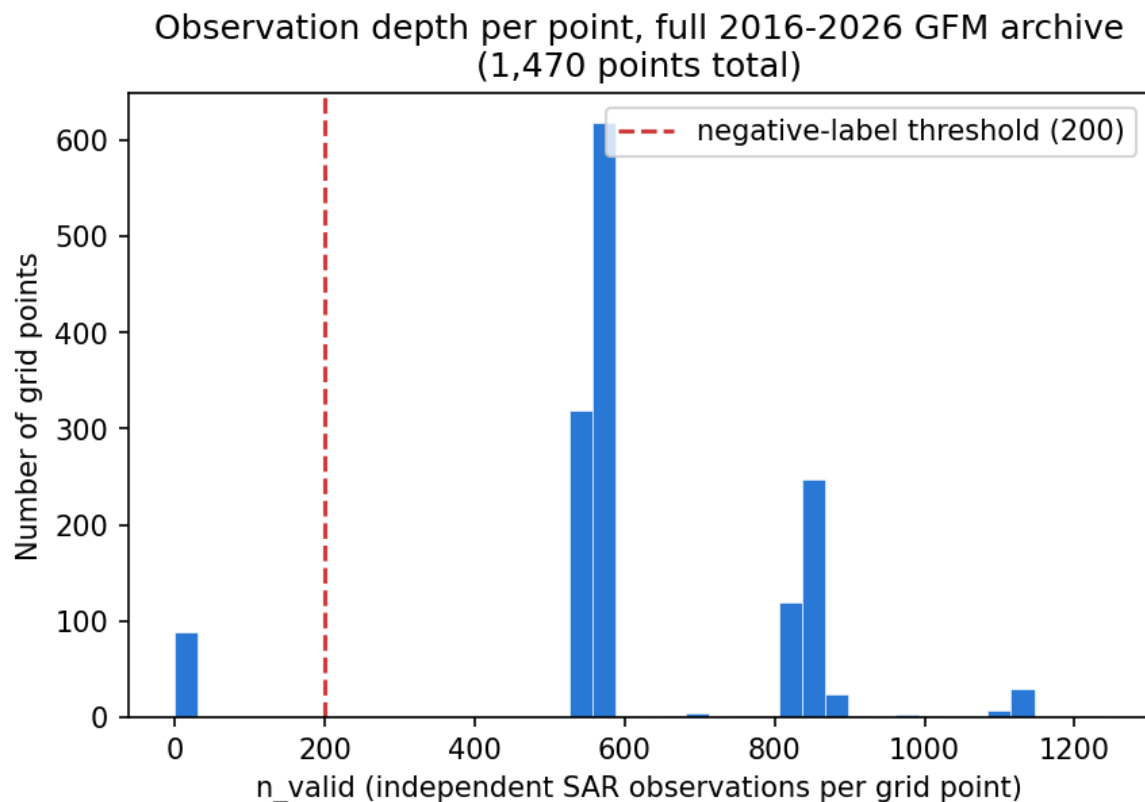


Figure 2. Observation depth per point — the full archive gives deep, consistent coverage, supporting the 200-observation negative-label threshold.

7.2 Model comparison & selection

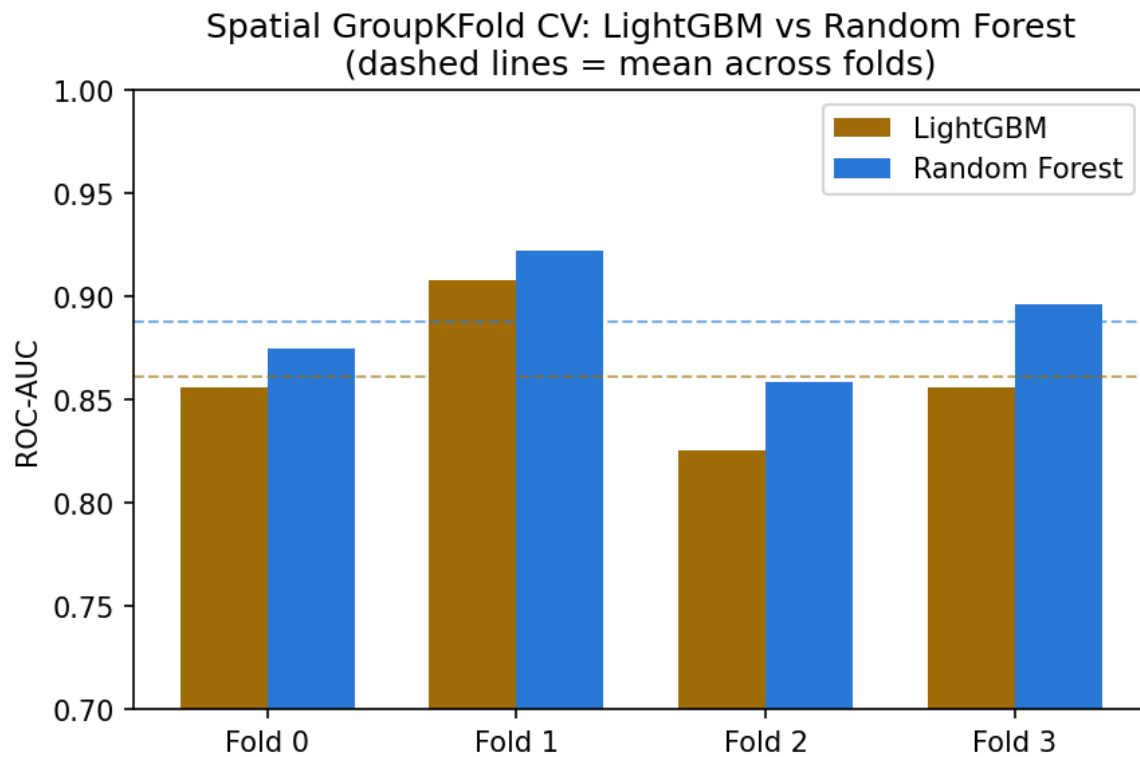


Figure 3. Spatial GroupKFold cross-validation: LightGBM vs. Random Forest, 4 folds. Random Forest wins on every fold.

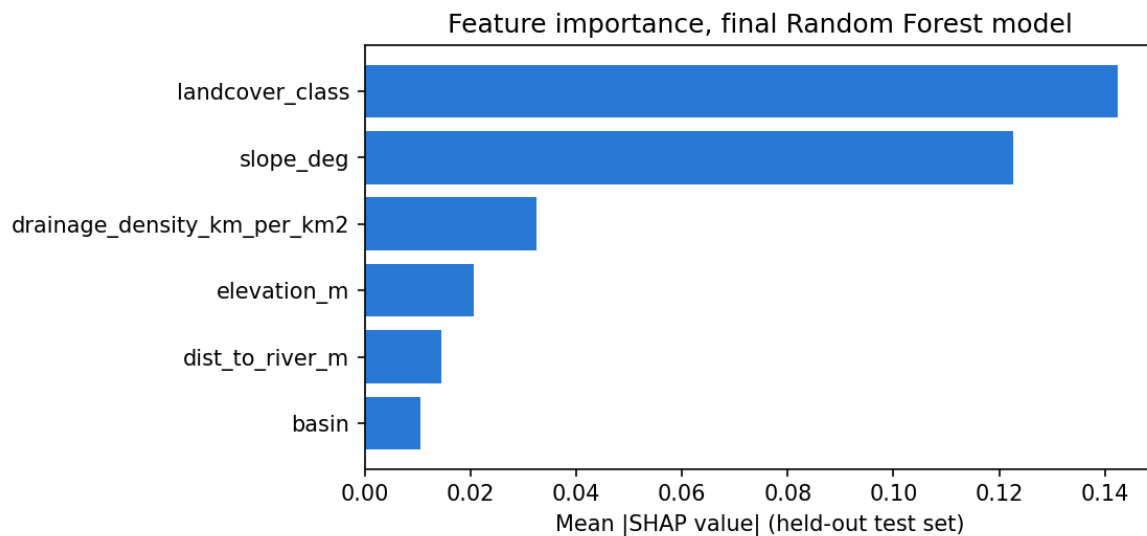


Figure 4. SHAP feature importance for the final Random Forest model on the held-out test set — land cover and slope dominate, consistent with flat deltaic terrain where these carry more signal than raw elevation.

7.3 Held-out spatial test performance

7 of 30 stations — stratified across all 4 basins so the test set still spans flat floodplain, haor wetlands, and hilly terrain — were held out completely: never touched during training or model selection.

Held-out spatial test set (7 stations, never touched during training/CV)

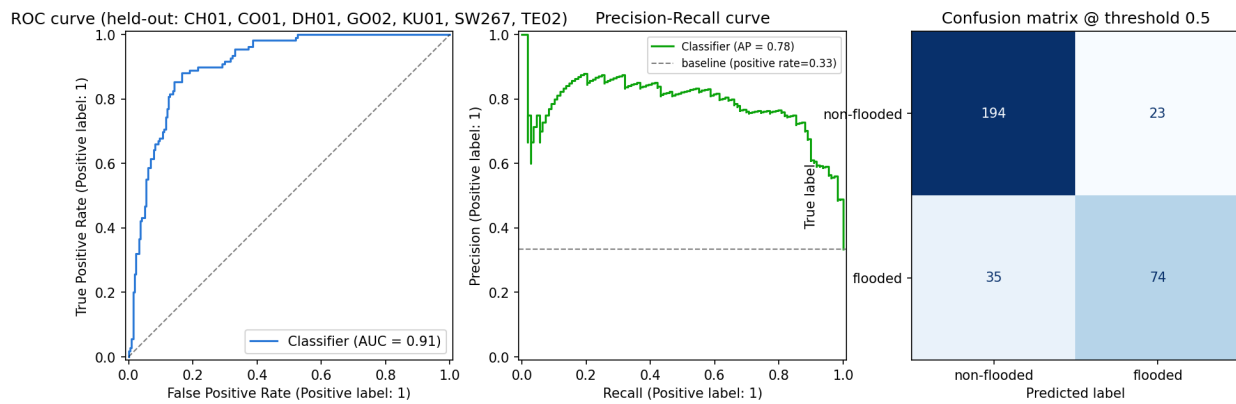


Figure 5. ROC curve, Precision-Recall curve, and confusion matrix (threshold 0.5) on the held-out test set.

7.4 Spatial & geographic patterns

30 monitored stations, colored by mean susceptibility score
(plain lat/lon scatter, not a GIS basemap)

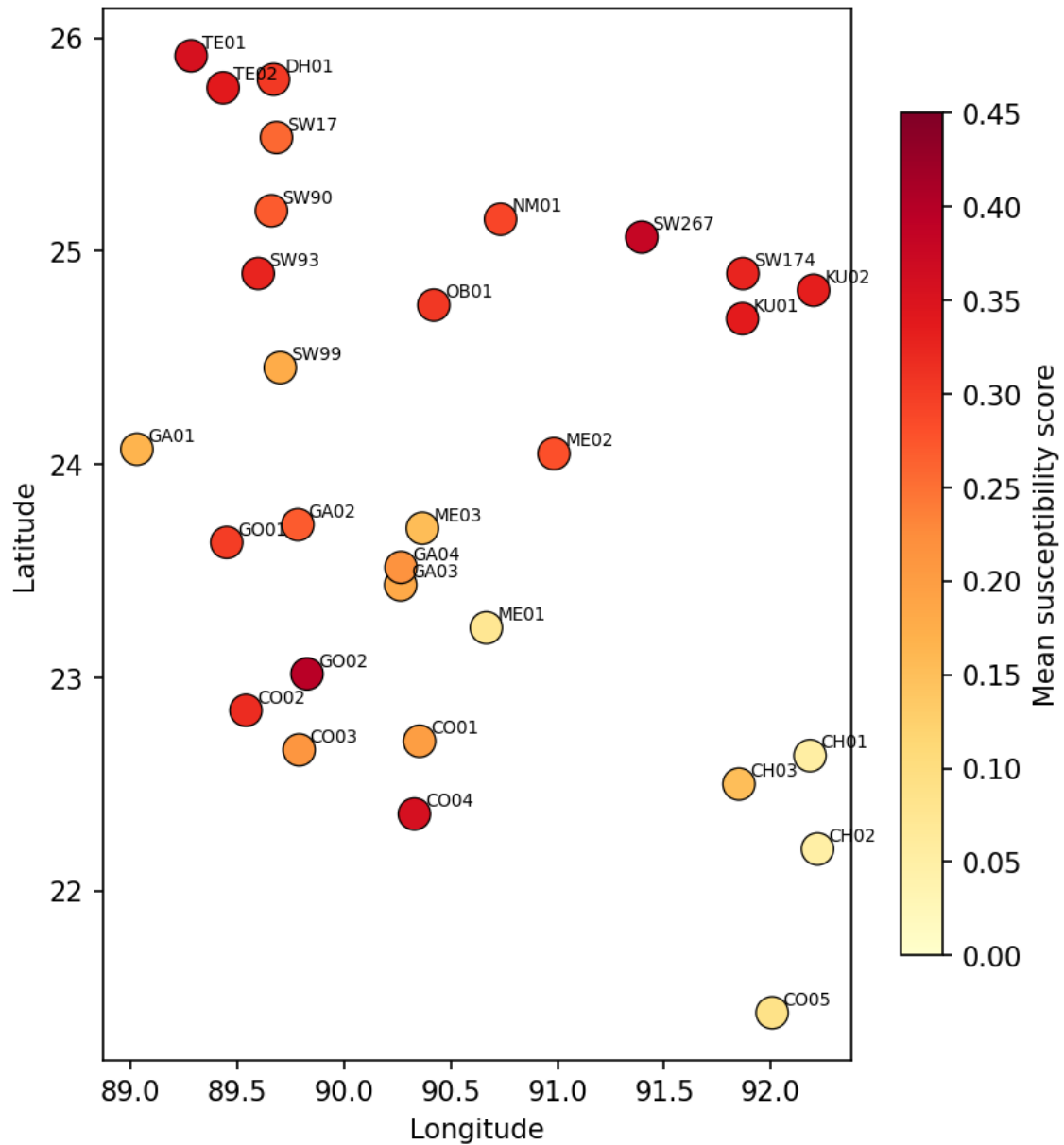


Figure 6. All 30 stations plotted by real coordinates, colored by mean susceptibility score. Meghna-basin haor/wetland stations (northeast) score highest; Chittagong Hill Tracts stations (southeast) score lowest — matching known Bangladesh flood geography.

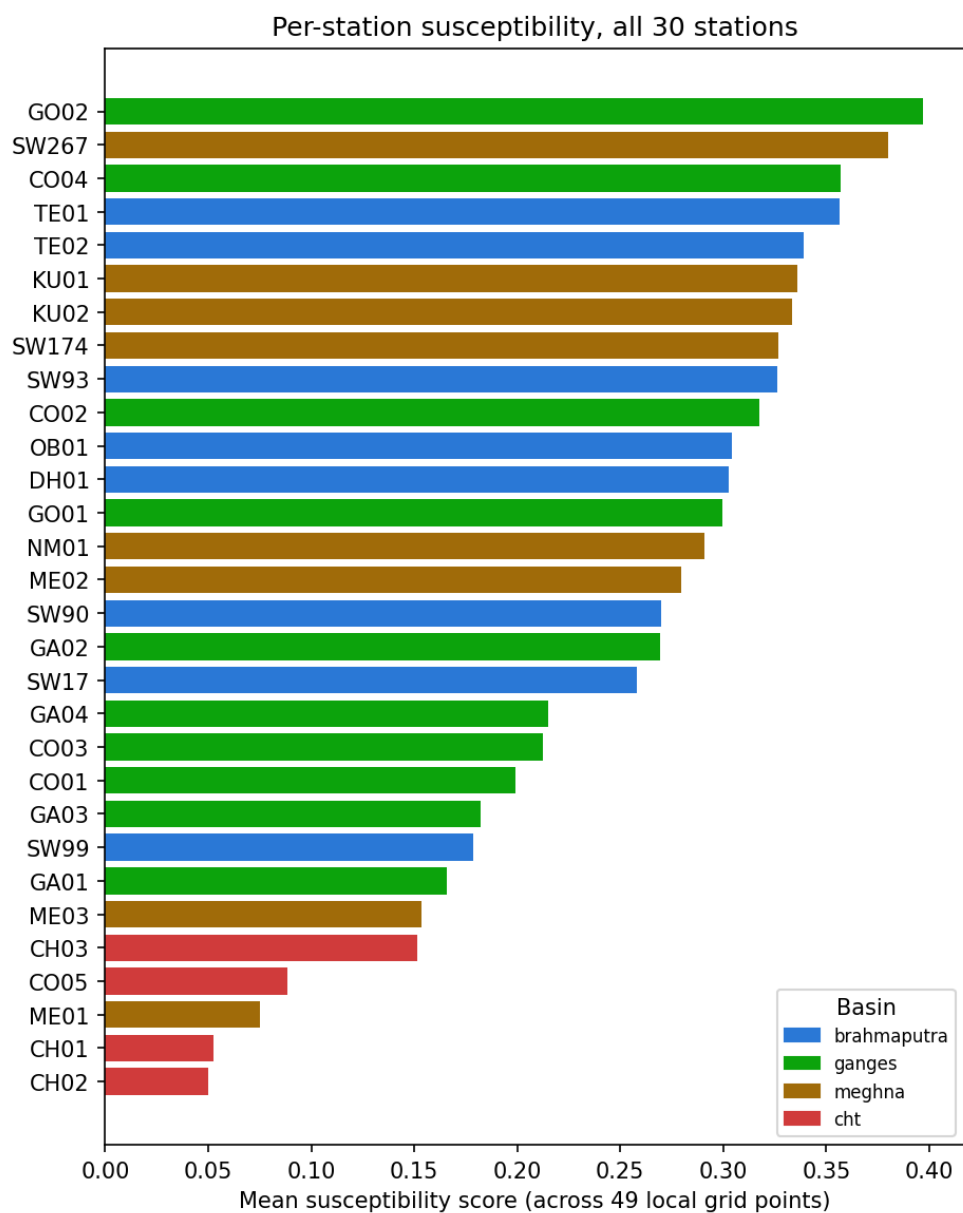


Figure 7. Per-station mean susceptibility score, all 30 stations, colored by basin.

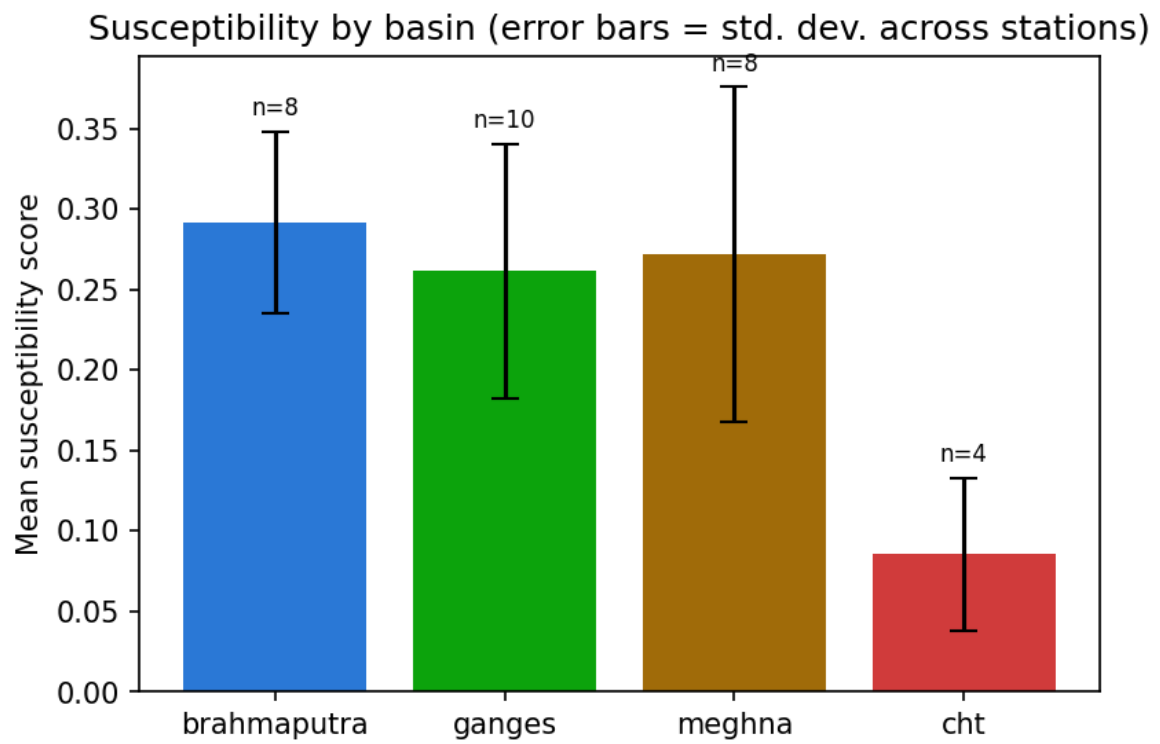


Figure 8. Mean susceptibility by basin (error bars = standard deviation across that basin's stations).

7.5 Land cover as a predictor

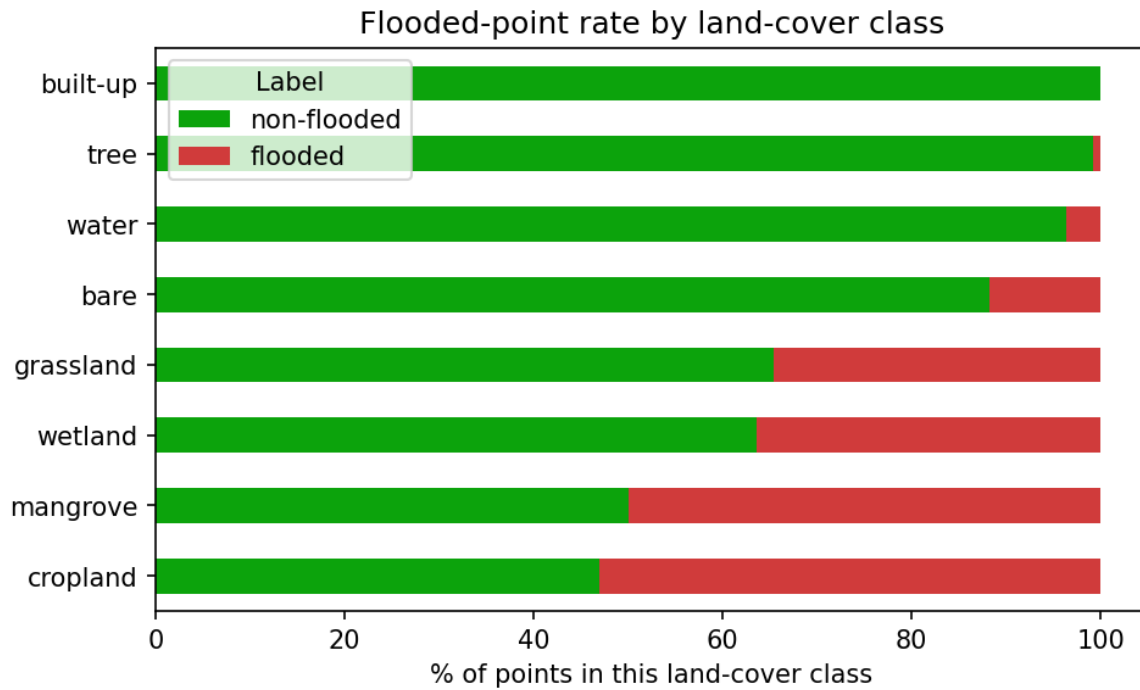


Figure 9. Flooded-point rate by land-cover class — water and wetland/mangrove classes show a far higher flooded rate than cropland or tree cover, matching physical intuition and explaining land cover's #1 SHAP ranking.

8. Comparison: This Model vs. the Literature

A direct accuracy comparison against the Bangladesh studies in §2.2 would be misleading without the methodology context below — presented explicitly rather than left implicit.

Study / model	Reported metric	Evaluation method	Comparable to this model?
Bangladesh comparative study (LightGBM/RF, §2.2)	97.7–97.8% accuracy	Not specified as spatial-holdout in the abstract; typical of the field is a random split	Not directly — likely benefits from the random-split inflation documented in §2.1
Sylhet AHP study (§2.2)	ROC $\approx$ 87%	Expert-weighted, non-ML	Partially — a non-inflated, realistic reference point
Northeast Bangladesh flash-flood RF study (§2.2)	RF beat XGBoost (exact AUC not extracted from search)	Not specified	Directly relevant methodologically — same RF-over-boosting finding reproduced independently here
This model (spatial GroupKFold CV)	0.888 mean ROC-AUC (RF), 0.862 (LightGBM)	Spatial group k-fold, grouped by station	—
This model (held-out spatial test)	0.908 ROC-AUC, 0.785 PR-AUC	7 stations held out completely, stratified across all 4 basins, never touched in training/CV	—

Why this model's 0.91 ROC-AUC is not directly beaten by the literature's 0.97+ accuracy figures, despite the lower headline number: accuracy and ROC-AUC are different metrics (accuracy can look high on an imbalanced set even with a mediocre model); and — more importantly — random-split evaluation on spatially autocorrelated data is documented to inflate reported AUC by 5–15% (§2.1). This model's 0.908 comes from a genuinely unseen set of stations, the harder and more honest evaluation standard. A fair apples-to-apples number is not available from the reviewed literature, because most reviewed studies do not report whether (or how) they controlled for spatial autocorrelation.

9. Integration with the Other Two Models

The three models combine via one transparent formula, deliberately chosen over a stacked meta-model so the combination stays defensible in one line rather than requiring its own separate justification:

$$\text{combined_risk} = \text{classifier_24h_probability} \times (0.5 + 0.5 \times \text{susceptibility_score})$$

Susceptibility acts as a bounded modulator (0.5–1.0×) on the classifier’s time-bound probability — it can reduce a forecast for inherently low-risk ground, but never fully zeroes it out, since an extreme enough storm can still flood low-susceptibility terrain. Implemented client-side in the dashboard (not inside either FastAPI service), since both model services are deliberately self-contained with zero dependency on each other or on this repository, so either can be copied to a different machine independently.

10. Limitations & Honesty Notes

- **Scope:** 30 station neighborhoods (1,470 points), not a national susceptibility atlas — deliberately scoped to what this project monitors, not a general-purpose Bangladesh map.
- **Land cover is a single snapshot** (ESA WorldCover 2021) — land use changes over time (e.g. new construction, deforestation) are not captured.
- **200-observation threshold is a real, inspected choice**, not arbitrary, but it is still a threshold: a small number of borderline points near that boundary could be re-labeled with a different (defensible) cutoff.
- **Drainage density remains self-derived** from a single DEM via standard flow-routing (§5.1) — not independently validated against ground survey data, though the underlying DEM itself (Copernicus GLO-30) is independently accuracy-assessed (§2.3). Distance-to-river no longer carries this caveat (§4.6).
- **HydroRIVERS is itself DEM-derived** (from a coarser ~500m HydroSHEDS DEM, not ground-surveyed) — an improvement in independence and provenance over the original self-derived feature, not an absolute ground-truth reference.
- **7-station held-out test is real but small** (326 points) — a larger held-out set would tighten the confidence interval around the 0.908 ROC-AUC figure.

11. Future Work

- **Done since the first version of this report:** mirrored the 3rd model card and combined-risk panel to the neumorphic frontend; cross-checked distance-to-river against an independent river-network layer and adopted the better-performing version (§4.6).
- **Blocked, not just low-priority:** re-scoring against a newer ESA WorldCover release — checked directly, no release newer than 2021 (v200) exists; an update is only “planned” with no date. Expanding the held-out test set as more stations are added is similarly blocked — there are no new stations to add yet.
- **Open, needs a design decision first:** whether a rainfall/soil-moisture dynamic modulator (rather than a fixed static score) could let susceptibility respond to antecedent wetness without turning it back into a second temporal model.

Appendix: Reproducibility Reference

Artifact	Path
Spatial grid definition	backend/train/susceptibility_grid.py
Label aggregation	backend/train/ingest_susceptibility_labels.py
Terrain/land-cover extraction	backend/train/ingest_susceptibility_terrain.py
Dataset merge + threshold decision	backend/train/build_susceptibility_dataset.py
Model training + spatial CV	backend/train/train_susceptibility_model.py
Distance-to-river independent cross-check	backend/train/crosscheck_distance_to_river.py
Distance-to-river variant comparison (§4.6)	backend/train/compare_river_distance_features.py
Trained model + metrics	backend/models/susceptibility/
Standalone service	packages/flood-susceptibility/
This report's figures	reports/generate_susceptibility_assets.py
Full build log	MODEL_BUILD_PLAN.md (2026-08-14 entries)